PVsyst V8.1.7

PVsyst - Simulation report

Grid-Connected System

Project: teresina

Variant: New simulation variant

No 3D scene defined, no shadings

System power: 2400 Wp

Teresina - Brazil

Author

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Project summary

Geographical Site
Teresina
Brazil

Situation

Latitude	-5.0875 °(S)
Longitude	-42.8050 °(W)
Altitude	72 m
Time zone	UTC-3

Project settings
Albedo 0.20

Weather data
Teresina
NASA POWER satellite data 2001-2020 - Synthetic

System summary

Grid-Connected System

No 3D scene defined, no shadings

Orientation #1
Horizontal plane

Near Shadings
no Shadings

User's needs
Unlimited load (grid)

System information

PV Array		Inverters	
Nb. of modules	4 units	Nb. of units	1 unit
Pnom total	2400 Wp	Total power	2 kWac
		Pnom ratio	1.20

Results summary

Simulation step	Hourly				
Produced Energy	4262.4 kWh/year	Specific production	1776 kWh/kWp/year	Perf. Ratio PR	83.46 %

Table of contents

Project and results summary 2
General parameters, PV Array Characteristics, System losses 3
Main results 4
Loss diagram 5
Predef. graphs 6
Single-line diagram 7

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16/10/26 16:46
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General parameters

Grid-Connected System — **No 3D scene defined, no shadings**

Simulation step
Hourly

Orientation #1
Horizontal plane

Models used

Transposition	Perez
Diffuse	Perez, Meteonorm
Circumsolar	separate

Horizon
Free Horizon

Near Shadings
no Shadings

User's needs
Unlimited load (grid)

PV Array Characteristics

PV module

Manufacturer	Generic
Model	JKM-600N-72HL4-V
(Original PVsyst database)	
Unit Nom. Power	600 Wp
Number of PV modules	4 units
Nominal (STC)	2400 Wp
Modules	1 unit x 4 In series
At operating cond. (50°C)	
Pmpp	2226 Wp
U mpp	163 V
I mpp	14 A

Inverter

Manufacturer	Generic
Model	MIC 2000TL-X
(Original PVsyst database)	
Unit Nom. Power	2.00 kWac
Number of inverters	1 unit
Total power	2.0 kWac
Operating voltage	50-500 V
Pnom ratio (DC:AC)	1.20

Total PV power

Nominal (STC)	2.40 kWp
Total	4 modules
Module area	10.3 m²
Cell area	9.5 m²

Total inverter power

Total power	2 kWac
Number of inverters	1 unit
Pnom ratio	1.20

Array losses

Thermal Loss factor
Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses
Global array res. 196 mΩ
Loss Fraction 1.50 % at STC

Module Quality Loss
Loss Fraction -0.75 %

Module mismatch losses
Loss Fraction 2.00 % at MPP

IAM loss factor
Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.963	0.892	0.814	0.679	0.438	0.000

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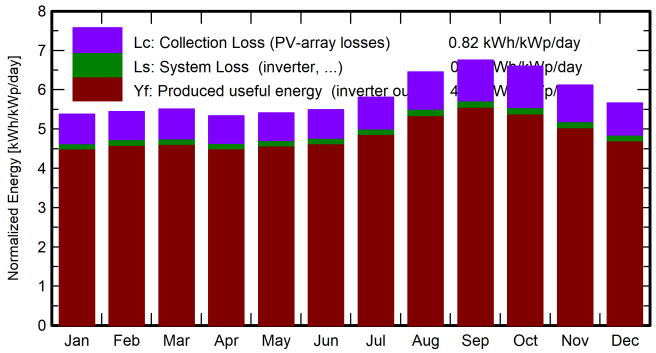
Main results

Simulation step
Hourly

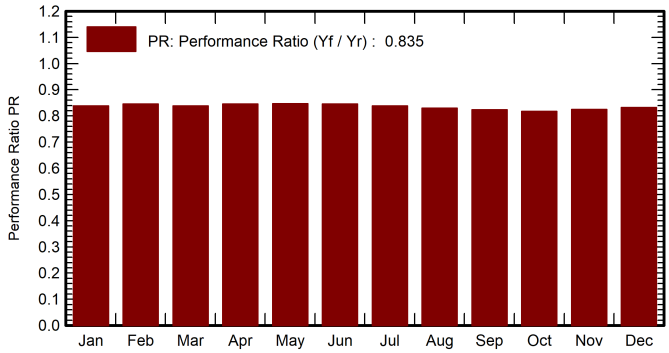
System Production

Produced Energy	4262.4 kWh/year	Specific production	1776 kWh/kWp/year
		Perf. Ratio PR	83.46 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor kWh/m²	DiffHor kWh/m²	T_Amb °C	GlobInc kWh/m²	GlobEff kWh/m²	EArray kWh	E_Grid kWh	PR ratio
January	166.7	76.94	27.35	166.6	162.3	345.1	335.0	0.838
February	152.4	71.82	26.30	152.4	148.7	318.3	309.0	0.845
March	170.6	78.40	26.15	170.6	166.3	353.9	343.5	0.839
April	160.0	68.62	25.91	160.0	156.0	334.4	324.6	0.845
May	167.6	59.46	26.32	167.6	163.3	351.0	340.7	0.847
June	164.7	45.65	26.86	164.7	160.4	344.0	333.9	0.845
July	179.9	42.61	27.82	179.9	175.4	373.0	362.1	0.839
August	200.1	43.64	29.21	200.1	195.8	410.4	398.5	0.830
September	202.6	50.59	30.69	202.5	198.4	412.2	400.3	0.824
October	204.6	61.86	31.29	204.6	200.6	413.5	401.1	0.817
November	183.5	67.00	30.81	183.5	179.5	374.3	363.2	0.825
December	175.6	74.54	29.58	175.5	171.3	361.1	350.4	0.832
Year	2128.5	741.13	28.20	2128.0	2077.9	4391.2	4262.4	0.835

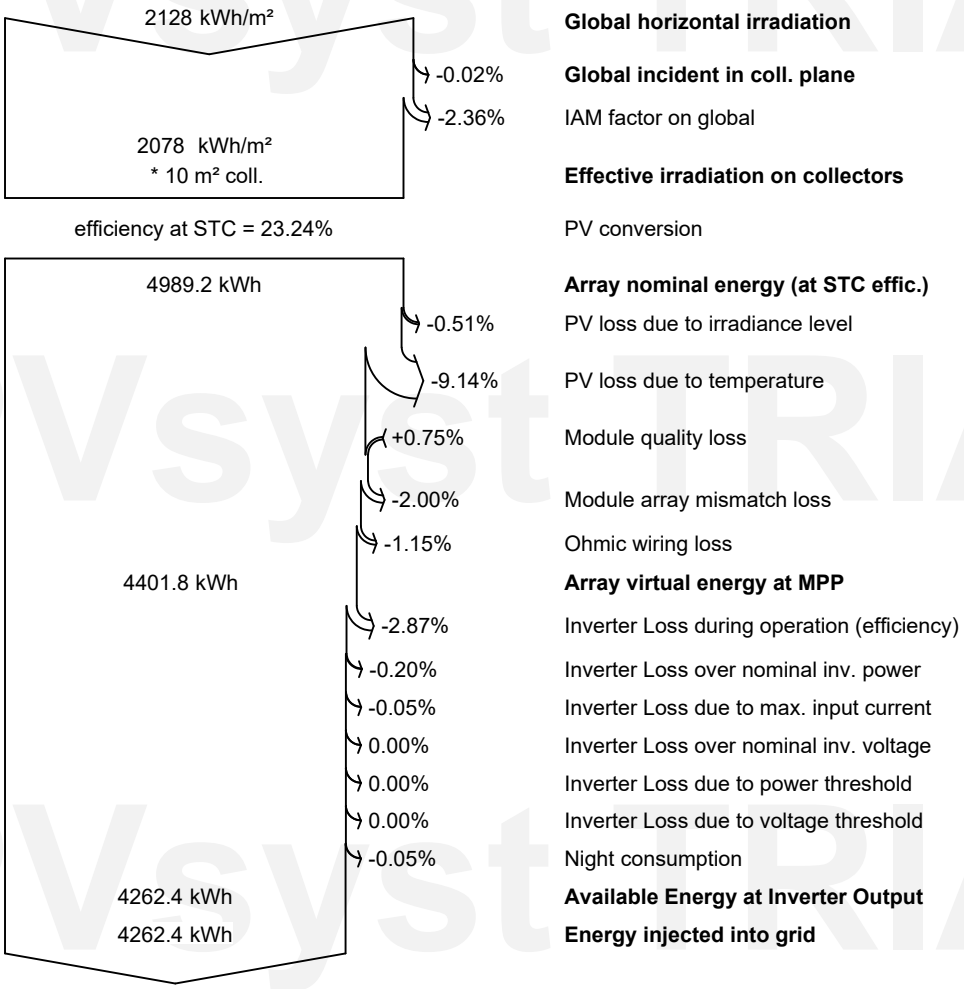
Legends

GlobHor	Global horizontal irradiation	EArray	Effective energy at the output of the array
DiffHor	Horizontal diffuse irradiation	E_Grid	Energy injected into grid
T_Amb	Ambient Temperature	PR	Performance Ratio
GlobInc	Global incident in coll. plane		
GlobEff	Effective Global, corr. for IAM and shadings		

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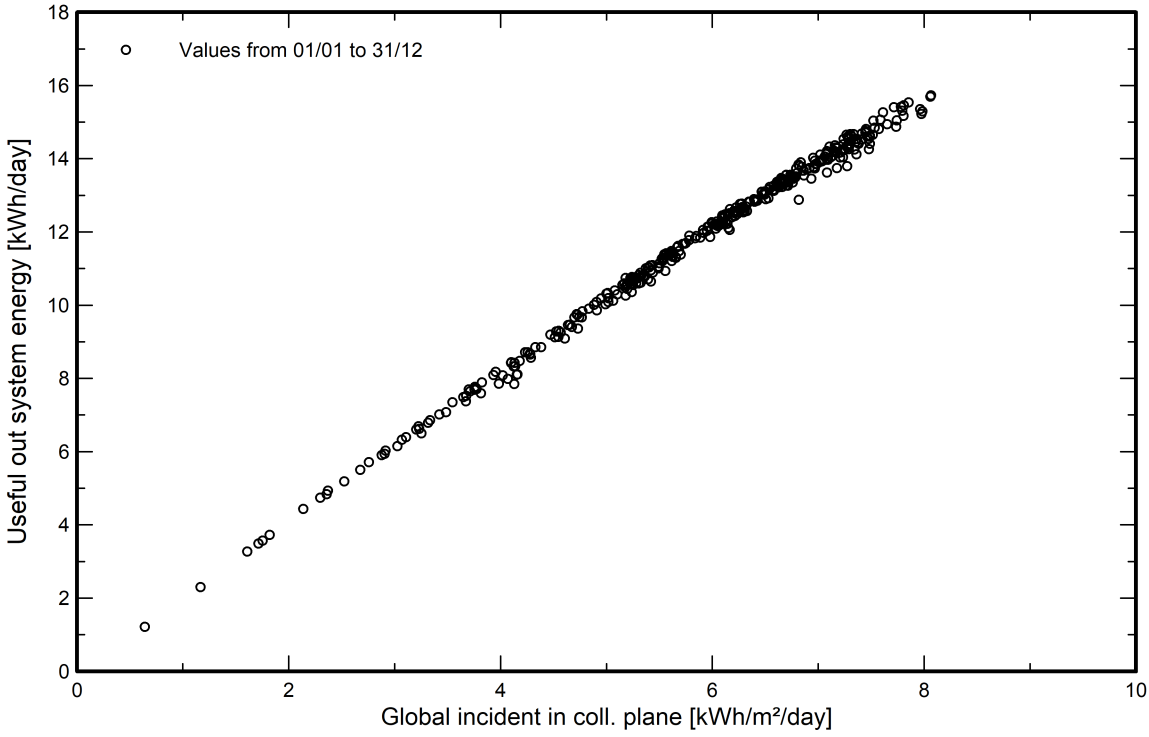
Loss diagram



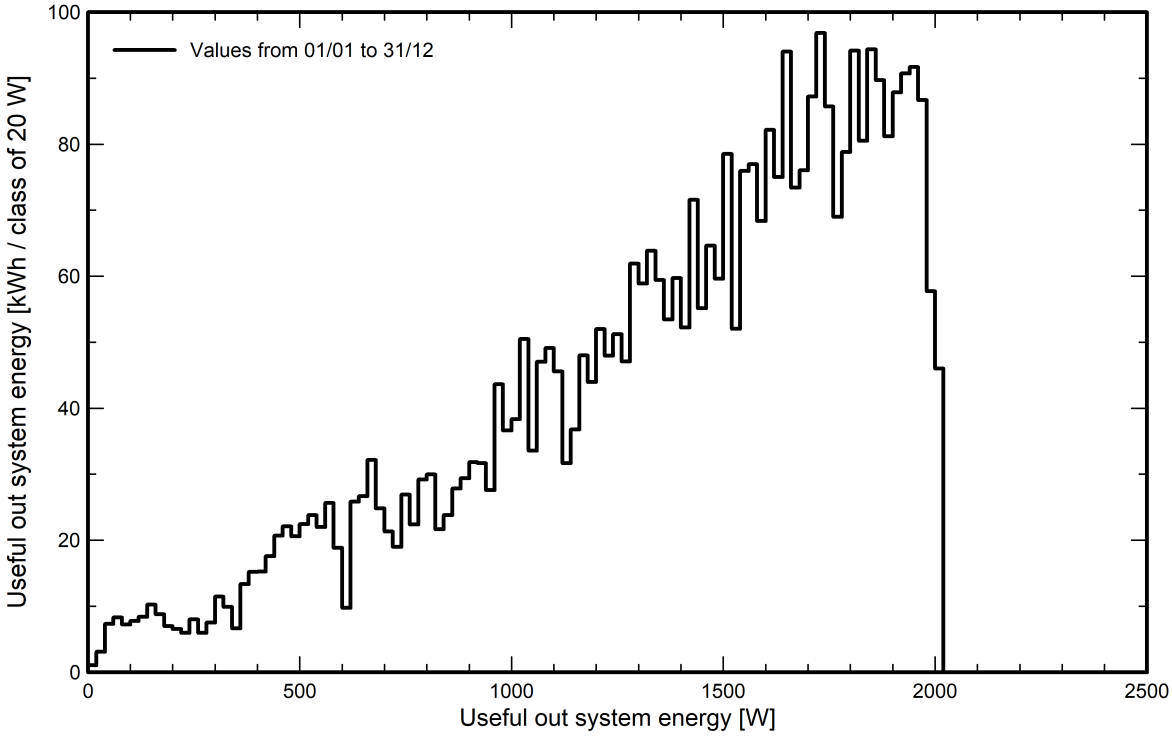
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Predef. graphs

Daily Input/Output diagram

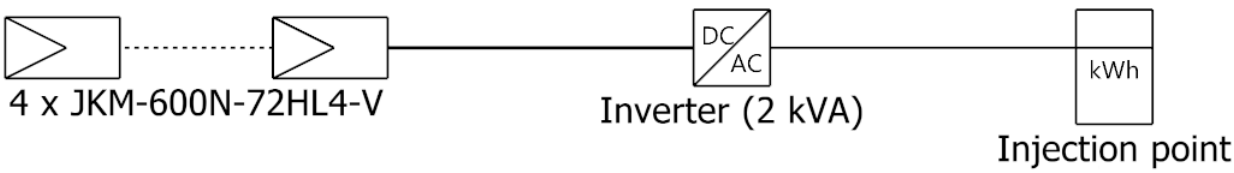


System Output Power Distribution



Single-line diagram

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PV module	JKM-600N-72HL4-V
Inverter	MIC 2000TL-X
String	4 x JKM-600N-72HL4-V

	teresina	
	VC0 : New simulation variant	16/10/26